QUANTUM THEORY AND THE ELECTRONIC STRUCTURE OF ATOMS

CHAPTER 1B

1. The Brackett series of the spectral line of atomic hydrogen appears in the infrared region. The wavelength for the third line in Brackett series is 2166 nm.

    a. Explain how the third line of the Brackett series is formed.
    b. Determine the energy involved to form the third line in the Brackett series.

2. An electron of a hydrogen atom is excited to the energy level n= 5 and drops to a lower energy level forming Paschen series.

    a. State the energy level whereby the electron 'dropped back'.
    b. Calculate the energy of the electron at the excited level.
    c. Calculate the energy emitted as a result of the transition.

3. One of the lines in the atomic hydrogen line emission spectrum falls in the visible light region and has a wavelength of 410 nm.

    a. Explain the meaning of line spectrum.
    b. Give the name for this spectrum series.
    c. Determine the transition of the electron involved.

4. One of the lines in the Balmer series of the atomic hydrogen emission spectrum has a wavelength of 486.2 nm. From which energy level does it drop from to give rise to this particular line?

5. Calculate the energy of a photon emitted by an electron in a Bohr atom moving from the 4^{th} to 1^{st} energy level.

6. Calculate the energy of a photon emitted by an electron in a Bohr atom going from the 6^{th} to 2^{nd} energy level.

7. Determine the wavelength and energy of radio waves with a frequency of 106.1 MHz

8. Calculate the frequency and energy of light with a wavelength of 500 nm.

9. The frequency and wavelength of light with energy 5 x 10^{-18} J.

10. Complete the following table:

Energy level, n	**Number of sublevels**	**Type of sublevel**	**Number of orbitals**	**Maximum number of electrons**	**Total number of electrons, $2n^2$**
4					
3					
2					
1					

11. Write the electron configuration of nitrogen atom using:

    a. orbital diagram
    b. *spdf* notation
    c. abbreviated electron configuration

12. Write the electron configuration of silicon using:

    a. orbital diagram
    b. abbreviated orbital diagram
    c. spdf notation
    d. abbreviated electron configuration

13. Write the electron configuration of the followings:

    a. Lithium
    b. Aluminium
    c. Fluorine
    d. Beryllium
    e. Iron

14. Write the electron configuration for the following ions:

    i. N^{3-}
    ii. Sc^{3+}
    iii. Mn^{2+}
    iv. Cl^{-}
    v. O^{2-}
    vi. Ca^{2+}
    vii. Fe^{2+}

15. Write the electron configuration of copper and chromium. Explain the abnormality.

16. Which of the following elements are paramagnetic? Which are diamagnetic?

    a. Potassium
    b. Zinc
    c. Fluorine

17. Give the values of *l* and *m* for orbitals 3p and 4d.

18. Determine the number of electrons in the indicated orbitals:

    a. 3d in zinc
    b. 2p in sodium
    c. 4p in arsenic
    d. 5s in rubidium

19. An element N has proton number of 22.

   a. Draw the orbital diagrams for the four outermost electrons in element N.
   b. Determine set of quantum number for all these four electrons.

20. Copper is the ninth element in the first row of d block of the Periodic Table.

   a. Write the electron configuration of copper according to the Aufbau principle.
   b. Write the actual configuration as determined by experiments.
   c. State the reason for any abnormality.
   d. Name another transition element which shows similar abnormality.
   e. Write the electron configuration of the element.